

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
--	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

departure airport name departure city arrival airport name arrival city airline name cost number status status

booking id: PK

[illegible]

flight_number	departure_airport_name	departure_city	arrival_airport_name	arrival_city	airline_name	ticket_price
KG101	Almaty Airport	Almaty	Astana Airport	Astana	Air Astana	50000

PK (booking_id and sear_number) as composite key

B001	12A	Ali Airman	P1001	KC101	Almaty Airport	Almaty	Astana Airport	Astana	Air Astana	30000
B001	12B	Ali	P1002	KC101	Almaty Airport	Almaty	Astana Airport	Astana	Air Astana	50000

passenger_passport_ passenger_full_name_

Possport number
PK

P1001	Ali Ahmad
-------	-----------

booking_id:
PK

booking_id	flight_number	departure_airport_name	departure_city	arrival_airport_name	arrival_city	airline_name	ticket_price
------------	---------------	------------------------	----------------	----------------------	--------------	--------------	--------------

B001	KC101	Almaty Airport	Almaty	Astana Airport	Astana	Air Astana	50000

PK(booking_id,
passport_number)
FK(booking_id &
passport_number)

booking_id	passenger_passport_number	seat_number
2001	21001	101

B001	P1001	12A
B001	P1002	12B

Airport_id
PK

PK(passport
_num)

Passport_num	full_name
P1001	Ali Ahmed

P1001	All Animals

airline id PK

airline_id	airline_name
1	Air Astana

AL001	Air Astana

flight_number	departure_airport_id	arrival_airport_id	airline_id
KO101	A001	A002	A001

KC101	A001	A002	AL001

fight_number: PK
depart_airport_id :FK
arrival_airport_id: FK
airline_id: FK

booking_id (PK)	flight_number (FK)	ticket_price
8881	US123	5000

B001	KC101	50000

booking_id (PK, FK)	passport_number (PK, FK)	seat_number (PK)
---------------------	--------------------------	------------------

B001	P1001	12A
B001	P1002	12B

pgAdmin 4

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Tables (5)

booking_receipt_1nf

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Preferences

postgres/postgres...

postgres/postgres@PostgreSQL 18*

postgres/postgres@PostgreSQL 18

No limit

history

Scratch Pad

Open in a new tab

1 DROP TABLE IF EXISTS booking_receipt_1nf;

2

3 CREATE TABLE booking_receipt_1nf (

4 booking_id VARCHAR(10),

5 passenger_full_name VARCHAR(100),

6 passenger_passport_number VARCHAR(20),

7 flight_number VARCHAR(20),

8 departure_airport_name VARCHAR(100),

9 departure_city VARCHAR(100),

10 arrival_airport_name VARCHAR(100),

11 arrival_city VARCHAR(100),

12 airline_name VARCHAR(100),

13 seat_number VARCHAR(10),

14 ticket_price NUMERIC(10, 2),

15

16 PRIMARY KEY (booking_id, seat_number)

17);

18

19 INSERT INTO booking_receipt_1nf (

Data Output

Messages

Notifications

Showing rows: 1 to 2

Page No: 1 of 1

	booking_id [PK] character varying (10)	passenger_full_name character varying (100)	passenger_passport_number character varying (20)	flight_number character varying (20)	departure_airport_name character varying (100)	departure_city character varying (100)	arrival_airport_name character varying (100)	arrival_city character varying (100)
1	B001	Ali Ahmad	P1001	KC101	Almaty Airport	Almaty	Astana Airport	Astana
2	B001	Ali Ahmad	P1001	KC101	Almaty Airport	Almaty	Astana Airport	Astana

Total rows: 2 Query complete 00:00:00.186

LF Ln 11, Col 31


```

1
2 create table Airline_info(
3     airline_id INT primary key,
4     airline_code varchar(30) not null,
5     airline_name varchar(50) not null,
6     airline_country varchar(50) not null,
7     created_at timestamp not null,
8     updated_at timestamp not null,
9     info varchar(50) not null
10 );
11 select * from airline;
12 create table Airport(
13     airport_id INT primary key,
14     airport_name varchar(50) not null,
15     country varchar(50) not null,
16     state varchar(50) not null,
17     city varchar(50) not null,
18     created_at timestamp not null,
19     updated_at timestamp not null
20 );
21 create table Baggage_check(
22     baggage_check_id INT primary key,

```

Data Output Messages Notifications












airline_id	airline_code	airline_name	airline_country	created_at	updated_at
[PK] integer	character varying (30)	character varying (50)	character varying (50)	timestamp without time zone	timestamp without time zone

Total rows: 0 Query complete 00:00:00.130

LF Ln 11, Col 1

pgAdmin 4

Object Explorer

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1.3 Sequences

Tables (20)

airline

airline1

airport

airport1

baggage

baggage_check

boarding_pass

booking

booking1

booking_flight

booking_passenger

contacts

flight

flight1

groups

passenger1

passengers

phonebook

phones

security_check

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Login/Group Roles

Tablespaces (2)

pg_default

pg_global

postgres/postgres@PostgreSQL 18

No limit

Query

Query History

Scratch Pad

```
22
23 create table Baggage_check(
24     baggage_check_id INT primary key,
25     check_resutl varchar(50) not null,
26     created_at timestamp not null,
27     updated_at timestamp not null,
28     booking_id int not null,
29     passenger_id int not null
30 );
31 select *from baggage_check;
32 create table Baggage(
33     baggage_id INT primary key,
34     weight_in_kg decimal(4, 2) not null,
35     created_at timestamp not null,
36     updated_at timestamp not null,
37     booking_id INT not null
38 );
39 create table Boarding_pass(
40     boarding_pass_id INT primary key,
41     booking_id INT not null,
42     seat varchar(50) not null,
43     boarding_time timestamp not null,
```

Data Output Messages Notifications

SQL

baggage_check_id	check_resutl	created_at	updated_at	booking_id	passenger_id
[PK] integer	character varying (50)	timestamp without time zone	timestamp without time zone	integer	integer

Total rows: 0 Query complete 00:00:00.074

Successfully run. Total query runtime: 74 msec. 0 rows affected.

LF Ln 31, Col 1

pgAdmin 4

Object Explorer

Object Explorer

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FTS Parsers

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1..3 Sequences

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airline

airline1

airport

airport1

baggage

baggage_check

boarding_pass

booking

booking1

booking_flight

booking_passenger

contacts

flight

flight1

groups

passenger1

passengers

phonebook

phones

security_check

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Login/Group Roles

Tablespaces (2)

pg_default

pg_global

postgres/postgres@PostgreSQL 18

No limit

Query

Query History

Scratch Pad

```
27 updated_at timestamp not null,
28 booking_id int not null,
29 passenger_id int not null
30 );
31 select *from baggage_check;
32 create table Baggage(
33     baggage_id INT primary key,
34     weight_in_kg decimal(4, 2) not null,
35     created_at timestamp not null,
36     updated_at timestamp not null,
37     booking_id INT not null
38 );
39 select *from baggage;
40 create table Boarding_pass(
41     boarding_pass_id INT primary key,
42     booking_id INT not null,
43     seat varchar(50) not null,
44     boarding_time timestamp not null,
45     created_at timestamp not null,
46     updated_at timestamp not null
47 );
48
```

Data Output Messages Notifications

baggage_id [PK] integer

weight_in_kg numeric (4,2)

created_at timestamp without time zone

updated_at timestamp without time zone

booking_id integer

Total rows: 0 Query complete 00:00:00.083

Successfully run. Total query runtime: 83 msec. 0 rows affected.

LF Ln 39, Col 1

pgAdmin 4

Object Explorer

FTS Dictionaries

FTS Parsers

FTS Templates

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Procedures

1.3 Sequences

Tables (20)

airline

airline1

airport

airport1

baggage

baggage_check

boarding_pass

booking

booking1

booking_flight

booking_passenger

contacts

flight

flight1

groups

passenger1

passengers

phonebook

phones

security_check

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Tablespaces (2)

pg_default

pg_global

Properties SQL Statistics Dependencies Dependents Processes Preferences postgres/postgres... postgres/postgres@PostgreSQL 18*

postgres/postgres@PostgreSQL 18

No limit

Query Query History

```
52 flight_id int not null,
53 created_at timestamp not null,
54 updated_at timestamp not null
55 );
56 select *from boarding_pass;
57 create table Booking(
58 booking_id Int primary key,
59 flight_id INT not null,
60 passenger_id INT not null,
61 booking_platform varchar(50) not null,
62 created_at timestamp not null,
63 updated_at timestamp not null,
64 status varchar(50) not null,
65 price decimal (7, 2) not null
66 );
67 select *from booking;
68 create table Flight(
69 flight_id INT primary key,
70 sch_departure_time timestamp not null,
71 sch_arrival_time timestamp not null,
72 departing_airport_id int not null,
73 arriving_airport_id int not null,
```

Scratch Pad

Data Output Messages Notifications

SQL

booking_id	flight_id	passenger_id	booking_platform	created_at	updated_at	status	ticket_price
[PK] integer	integer	integer	character varying (50)	timestamp without time zone	timestamp without time zone	character varying (50)	numeric (7,2)

Total rows: 0 Query complete 00:00:00.073

LF Ln 67, Col 1

pgAdmin 4

Object Explorer

FTS Dictionaries

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FTS Templates

Foreign Tables

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Procedures

1.3 Sequences

Tables (20)

airline

airline1

airport

airport1

baggage

baggage_check

boarding_pass

booking

booking1

booking_flight

booking_passenger

contacts

flight

flight1

groups

passenger1

passengers

phonebook

phones

security_check

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Login/Group Roles

Tablespaces (2)

pg_default

pg_global

postgres/postgres@PostgreSQL 18

No limit

Query

Query History

Scratch Pad

```
64      status varchar(50) not null,
65      price decimal (7, 2) not null
66  );
67  select *from booking;
68  create table Flight(
69      flight_id INT primary key,
70      sch_departure_time timestamp not null,
71      sch_arrival_time timestamp not null,
72      departing_airport_id int not null,
73      arriving_airport_id int not null,
74      departing_gate varchar(50) not null,
75      arriving_gate varchar(50) not null,
76      airline_id Int not null,
77      act_departure_time timestamp not null,
78      act_arrival_time timestamp not null,
79      created_at timestamp not null,
80      updated_at timestamp not null
81  );
82  select *from flight;
83  create table Passengers(
84      passenger_id int primary key,
85      first_name varchar(50) not null,
```

Data Output Messages Notifications

sch_departure_time

timestamp without time zone

sch_arrival_time

timestamp without time zone

departing_airport_id

integer

arriving_airport_id

integer

departing_gate

text

arriving_gate

character varying (50)

airline_id

integer

act_departure_time

timestamp without time zone

act_arrival_time

timestamp without time zone

Total rows: 0 Query complete 00:00:00.071

LF Ln 82, Col 1

pgAdmin 4

Object Explorer

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1.3 Sequences

Tables (20)

airline

airline1

airport

airport1

baggage

baggage_check

boarding_pass

booking

booking1

booking_flight

booking_passenger

contacts

flight

flight1

groups

passenger1

passengers

phonebook

phones

security_check

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Login/Group Roles

Tablespaces (2)

pg_default

pg_global

postgres/postgres@PostgreSQL 18

No limit

Query

Query History

Scratch Pad

```
79      created_at timestamp not null,  
80      updated_at timestamp not null  
81  );  
82  select *from flight;  
83  create table Passengers(  
84      passenger_id int primary key,  
85      first_name varchar(50) not null,  
86      last_name varchar (50) not null,  
87      date_of_birth date not null,  
88      gender varchar(50) not null,  
89      country_of_citizenship varchar (50) not null,  
90      country_of_residence varchar (50) not null,  
91      passport_number varchar(20) not null,  
92      created_at timestamp not null,  
93      updated_at timestamp not null  
94  );  
95  select *from passengers;  
96  create table Security_check(  
97      security_check_id int primary key,  
98      check_result varchar(20) not null,  
99      created_at timestamp not null,  
100     updated_at timestamp not null,
```

Data Output Messages Notifications

passenger_id [PK] integer

first_name character varying (50)

last_name character varying (50)

date_of_birth date

gender character varying (50)

country_of_citizenship character varying (50)

country_of_residence character varying (50)

passport_number character varying (20)

created_at timestamp without time zone

Total rows: 0 Query complete 00:00:00.073

LF Ln 95, Col 1

pgAdmin 4

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1..3 Sequences

Tables (20)

airline

airline1

airport

airport1

baggage

baggage_check

boarding_pass

booking

booking1

booking_flight

booking_passenger

contacts

flight

flight1

groups

passenger1

passengers

phonebook

phones

security_check

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Tablespaces (2)

pg_default

pg_global

postgres/postgres@PostgreSQL 18

No limit

Query

Query History

Scratch Pad

```
94 );
95 select *from passengers;
96 create table Security_check(
97     security_check_id int primary key,
98     check_result varchar(20) not null,
99     created_at timestamp not null,
100     updated_at timestamp not null,
101     passenger_id int not null
102 );
103 select *from security_check;
104 alter table airline_info
105 rename to airline;
106 alter table booking
107 rename column price to ticket_price;
108 alter table flight
109 alter column departing_gate type text;
110 alter table airline
111 drop column info;
112
113 alter table Security_check
114 add constraint fk_security_passenger
115 foreign key (passenger_id)
```

Data Output Messages Notifications

SQL

security_check_id [PK] integer	check_result character varying (20)	created_at timestamp without time zone	updated_at timestamp without time zone	passenger_id integer
-----------------------------------	--	---	---	-------------------------

Total rows: 0 Query complete 00:00:00.087 LF Ln 103, Col 1

pgAdmin 4

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Tables (20)

airline

airline1

airport

airport1

baggage

baggage_check

boarding_pass

booking

booking1

booking_flight

booking_passenger

contacts

flight

flight1

groups

passenger1

passengers

phonebook

phones

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Tablespaces (2)

pg_default

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No limit

Query

Query History

Scratch Pad

```
103 select * from security_check;
104 alter table airline_info
105 rename to airline;
106 alter table booking
107 rename column price to ticket_price;
108 alter table flight
109 alter column departing_gate type text;
110 alter table airline
111 drop column info;
112
113 alter table Security_check
114 add constraint fk_security_passenger
115 foreign key (passenger_id)
116 references passengers(passenger_id);
117
118 alter table Booking
119 add constraint fk_booking_passenger
120 foreign key (passenger_id)
121 references passengers(passenger_id);
122
123 alter table Baggage_check
124 add constraint fk_baggage_check_id
125 foreign key (passenger_id)
126 references passengers(passenger_id);
127
128 ALTER TABLE baggage_check
129 ADD CONSTRAINT fk_baggage_check_booking
130 FOREIGN KEY (booking_id)
131 REFERENCES booking(booking_id);
132
133 ALTER TABLE baggage
134 ADD CONSTRAINT fk_baggage_booking
135 FOREIGN KEY (booking_id)
136 REFERENCES booking(booking_id);
137
138 ALTER TABLE ...
```

Data Output Messages Notifications

security_check_id

check_result

created_at

updated_at

passenger_id

Total rows: 0 Query complete 00:00:00.087

LF Ln 103, Col 1

pgAdmin 4

Object Explorer

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airline

airline1

airport

airport1

baggage

baggage_check

boarding_pass

booking

booking1

booking_flight

booking_passenger

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Query History

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161

162

163

ADD CONSTRAINT fk_baggage_check_booking

FOREIGN KEY (booking_id)

REFERENCES booking(booking_id);

ALTER TABLE baggage

ADD CONSTRAINT fk_baggage_booking

FOREIGN KEY (booking_id)

REFERENCES booking(booking_id);

ALTER TABLE boarding_pass

ADD CONSTRAINT fk_boarding_pass_booking

FOREIGN KEY (booking_id)

REFERENCES booking(booking_id);

ALTER TABLE booking_flight

ADD CONSTRAINT fk_booking_flight_booking

FOREIGN KEY (booking_id)

REFERENCES booking(booking_id);

ALTER TABLE flights

ADD CONSTRAINT fk_flight_departure_airport

FOREIGN KEY (departing_airport_id)

REFERENCES airport(airport_id);

ALTER TABLE flights

ADD CONSTRAINT fk_flight_arrival_airport

FOREIGN KEY (arriving_airport_id)

REFERENCES airport(airport_id);

ALTER TABLE flights

ADD CONSTRAINT fk_flight_airline

FOREIGN KEY (airline_id)

REFERENCES airline(airline_id);

Data Output

Messages

Notifications

SQL

security_check_id

check_result

created_at

updated_at

passenger_id

[PK] integer

character varying (20)

timestamp without time zone

timestamp without time zone

integer

Total rows: 0 Query complete 00:00:00.087

LF Ln 103, Col 1

Normalization part:

3. Unnormalized functional dependency: booking_id to all non-key attributes, except passenger name, passport, and set number.

1NF: the full composite key to all of non-key attributes, booking_id alone functional dependency to flight number, departure (airport, city), arrival (city, airport), airline name, and ticket_price. Passport number can determine passenger name so they also have functional dependency.

2NF: functional dependencies: passport number to passenger full name. In the booking table booking_id to all other non-key attributes. And, booking_id and passport_number determines both determines seat number

3NF: functional dependency: passport_number to full name. airline_id to airline name. booking_id to flight number and ticket_price. Airport_id to airport_name and city. Fight_number to departure and arrivall airport_id, and airline_id. Booking_id and passport_number to seat_number.

4. The Flights table is in 3NF. Its primary key is flight_id, which uniquely identifies each flight. The main functional dependency is flight_id → all other attributes in the table, including scheduled and actual departure/arrival times, airport IDs, gates, airline ID, and timestamps. airline_id determines airline information, while departing_airport_id and arriving_airport_id determine airport information, but these details are stored in separate Airline and Airport tables. Since attributes such as airline_name or airport_name are not stored in the Flights table, there is no transitive dependency between non-key attributes in this table. Therefore, the Flights table satisfies 3NF.